

Funding a Venture

Prof. Dr. Orestis Terzidis



INTRODUCTION BUSINESS PLANNING

Why have a business plan?

- Every **major project** needs a business plan. It is a **roadmap** and basis for funding decisions.
- Writing a business plan is an **intensely focused activity**. It requires **honest thinking** about your business concept.
- The more mature a business, the more you need a structured plan. Do not over-plan in the first phases.
- Steve Blank: “No Business Plan Survives the First Contact With A Customer”.



The Plan Depends on the Maturity

Start-up Life cycle



The **more mature** a business is, **the more we rely on numbers**.
For a **newer business**, the numbers matter less and the **words matter more**.

Robert Mahoney, Investment Banker
(Cf. Cristea, 2016)

What is your purpose?

What is your context?

- Resource-rich big corporation -> reduced marketing and operations plan
- Resource constrained start-up -> opportunity, financial plan, team, competition

Who is your audience?

- Management committee: opportunity within the corporate context
- Lenders: risk, payback period, cash flow
- Investors: risks, breakeven point, long-term value



EXIST Program (initiated in 1998)

Key objectives of EXIST

- Establish a culture of entrepreneurship in university teaching, research and administration [...].
- [...] translate the findings of academic research into economic value [...].
- Promote the huge potential for business ideas and entrepreneurs at universities and research institutions [...].
- [...] increase the number of innovative business start-ups and create [...] new jobs [...]

Key Instruments of EXIST

- Entrepreneurship Culture
- Start-up Grant
- Research Transfer



EXIST Start-Up Grant

- Supports the **preparation of innovative business start-up** projects at universities [...].
 - The grant aims to help scientists, [...] graduates and students to develop their [...] ideas and create **business plans**, [...] **products & services**.
 - The grant covers **living expenses** of the founders. They receive a grant between **1000 to 3000 €/month**, depending on their degree, for a maximum period of 12 months.
- In addition, they receive
- **Materials and equipment** (worth 10,000-17,000€),
 - **Funding for coaching** (5,000 euro),
 - **Child benefit** of 100 euro per month and child.
 - The university [...] offers them **infrastructure** during the pre-start-up phase and provides technical and start-up-related assistance.

ANATOMY OF A BUSINESS PLAN

Typical structure of a business plan

- Cover page and table of content
- Executive summary
- Business description
- Business environment analysis
- Industry background
- Competitive analysis
- Market analysis
- Marketing plan
- Operations plan
- Management team
- Financial plan
- Attachments and milestones



Exist Start-up Grant Application

EXIST Grant Application Template

<http://www.exist.de/DE/Programm/Exist-Gruenderstipendium/Projektantrag/inhalt.html>

1. Executive Summary

2. Business Idea

2.1. Startup History

2.2 Expertise

2.3. Innovation

2.4. Project Planning

3. Market/Competition

3.1. Market situation

3.2 USP & Value Proposition

3.3 Competitors

3.4. Market Entry

4. Operational Planning

4.1. Financial Planning

4.2. Firm Organization

4.3. Opportunities & Risks

0| Cover page & table of content

- The first impression counts.
- Have a clean, professional appearance.
- Include the name of the business or project.
- Include the logo.
- Make sure you have a complete table of content.



1| Executive summary

EXIST recommendations

- Problem & solution
- Management team
- Industry and market environment
- Competition
- Business Model

Potential further points

- Financial potential
- Resources or capital requested
- Formal facts about the company
- Mission statement



Despite the variations in the structure of business plans, all [...] comprise descriptions of the opportunity, the context, the managers, and the financial risk and reward.

Cyr (2007)



2| Business Idea

2.1. Stage & History

- Author(s) of the business idea
- Existing IP rights
- Link to previous projects
- Integration in the university

2.2 Experts

- Team and distribution of tasks
- Scientific work related to the project
- Entrepreneurship education & business experience
- Partners & mentors



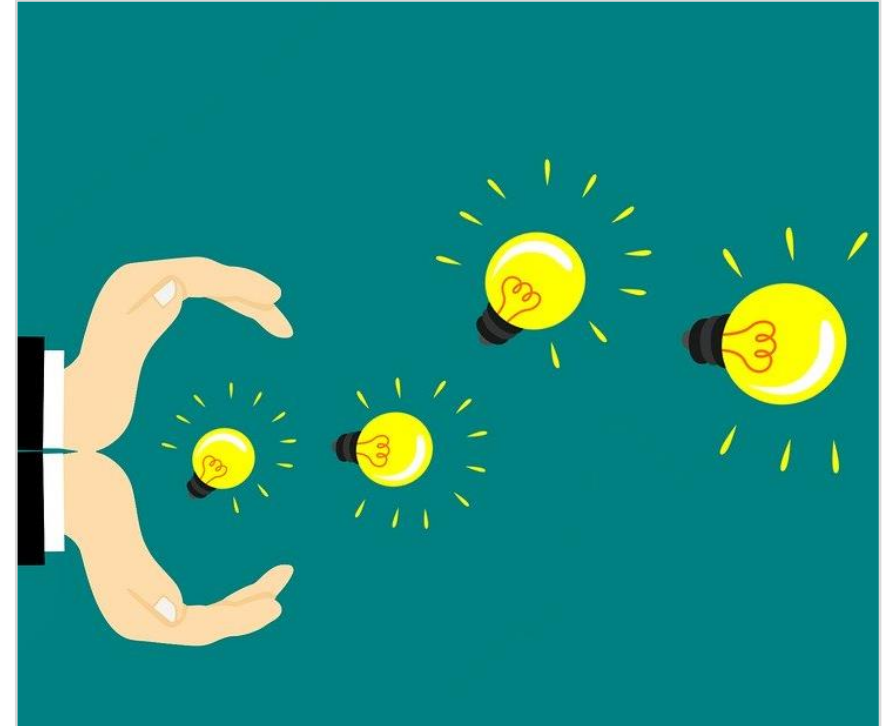
2| Business Idea

2.3. Innovation

- Description of the technology or product/service innovation
- Development plan and status of implementation
- Technology readiness level

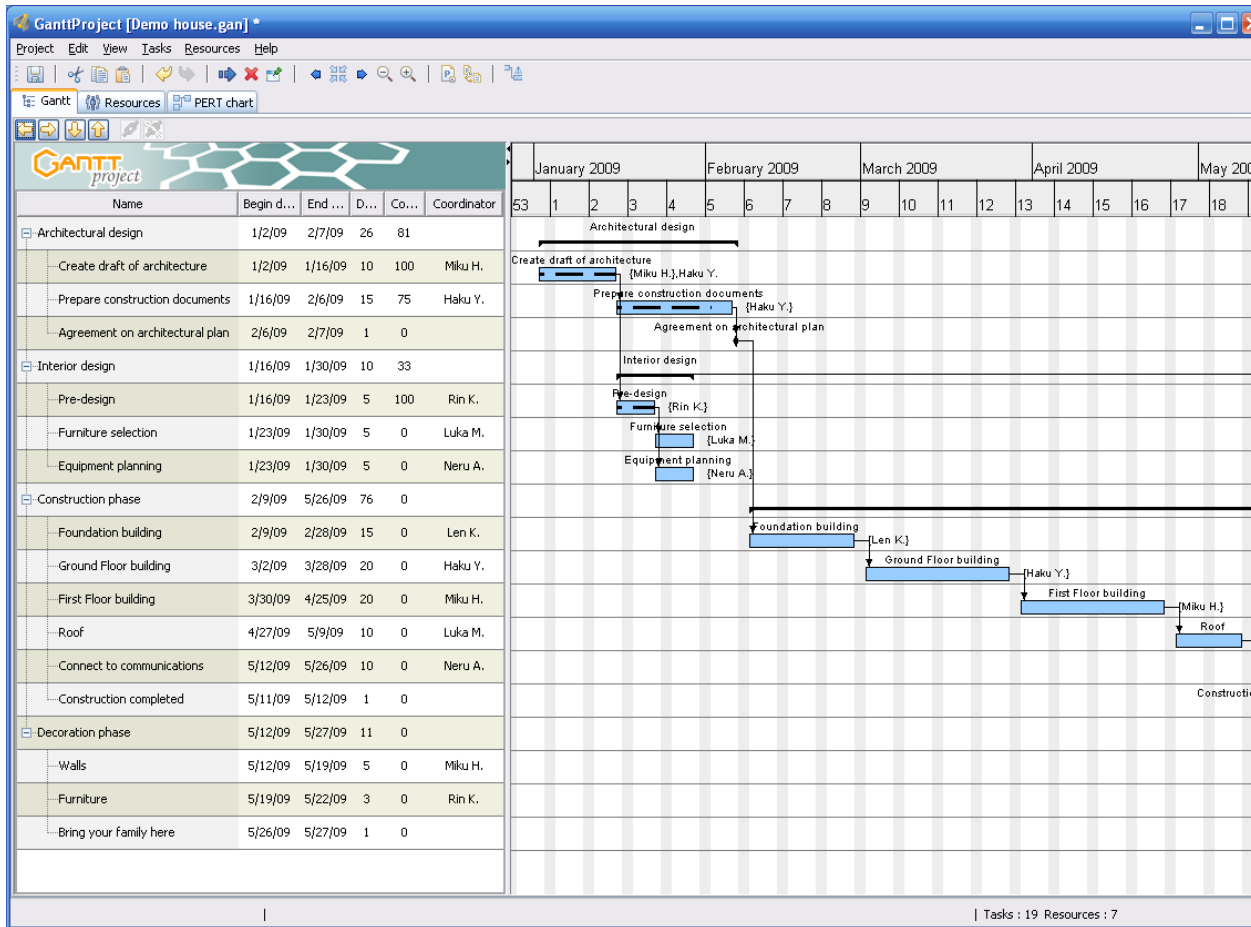
2.4. Project Planning

- Project work breakdown structure for the grant period
- Outlook on activities until market entry or next financing round



Gantt Chart for building a house

Source: wikipedia



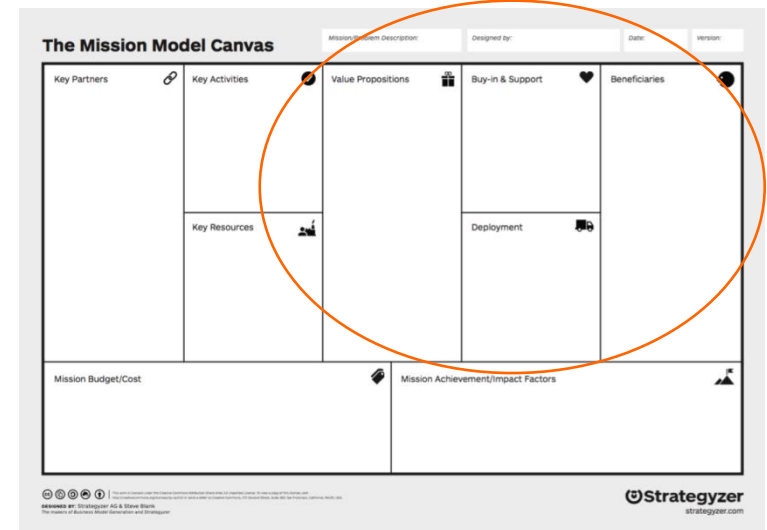
3| Market & Competition

3.1. Market Situation

- Market segments
- Data on market size
- Forecast market growth
- Market potential
- Industry Analysis

3.2 USP and customer value

- Customer value proposition of the product or service
- Unique feature of the product or service over competitors

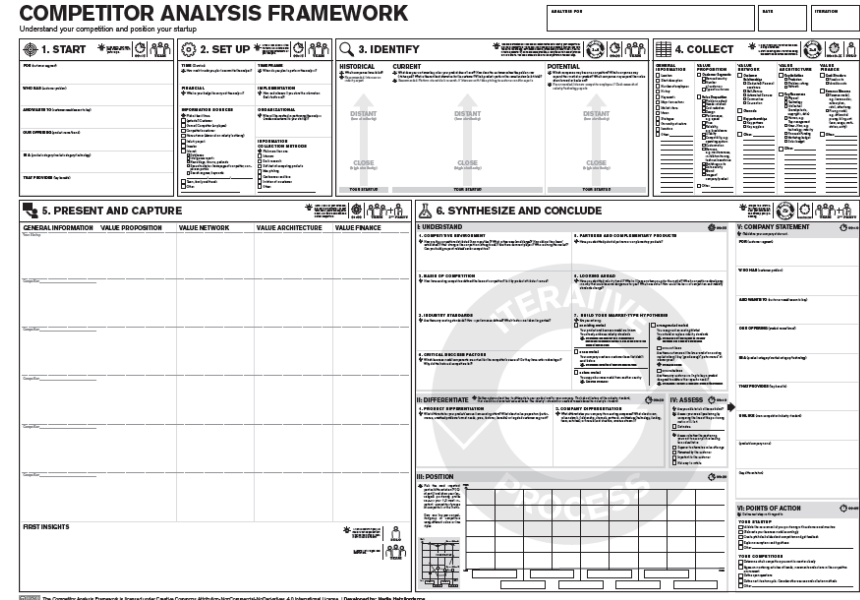


3| Market & Competition

3.3 Competitors

- Significant competitors
- Differentiation with respect to their offers
- You can use a tool developed in a doctoral thesis at the KIT: Competitor Analysis Framework (CAF)

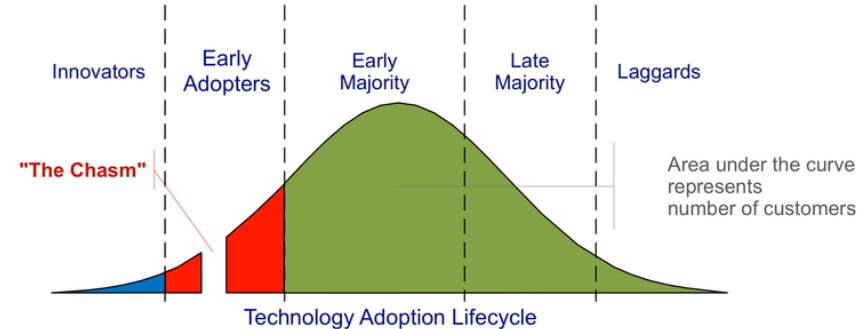
Hatzijordanou (2019 a, b)



3| Market & Competition

3.4. Market Entry

- Details of the first target group
- Potential pilot customers (LoI)
- Possible market entry barriers
- Concrete marketing & sales activities
- Strategic partnerships for market entry



4| Operations Plan

4.1. Financial Plan

- Revenue and expenditure forecasts for the first two years with explanation
- Outline of the proposed pricing model
- Define revenue streams (e.g. software licenses & consulting)
- Define expenditures (e.g. salaries, infrastructure, cloud capacity, license fees)

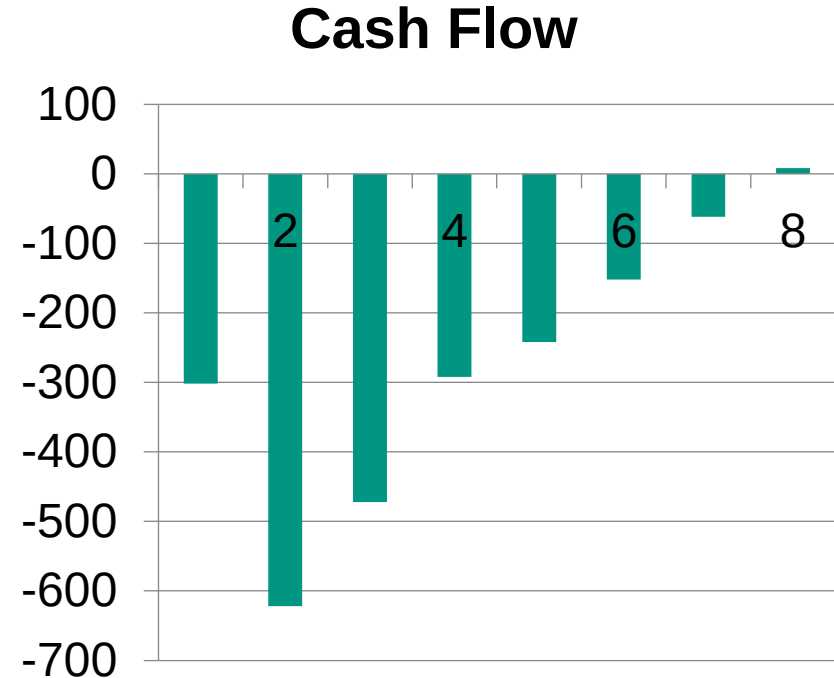
	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
Rev1	...	...						
Rev2								
Exp1								
Exp2								
Exp3								
Sum	...	...	...					

Period	1	2	3	4	5	6	7	8
Revenues								
Orders		100	180	190	210	230	250	270
Invoice				100	180	190	210	230
Payment							100	180
Cost								
Personnel	50	50	50	50	50	50	50	50
Goods	10	30	50	40	140	60	70	80
Other	40	40	70	60	50	40	40	40
Interest & Tax	2	2	2	2	2	2	2	2
Investments								
	200	500	300	140				
Liquidity								
Cash Flow	-302	-622	-472	-292	-242	-152	-62	+8
Accumulated	-302	-924	-1396	-1688	-1930	-2082	-2144	-2136

Cash Flow Plan

- If you wanted to follow the plan as laid out, you need at least **2144 k€**.
- In particular, you need to finance the time between order and payment (**working capital**).
- In addition, you need to pre-finance **investments**.
- The cash flow table does not contain depreciations.

Cash is King!



Pricing

- The price is a key factor for **profits**. (Simon 2004)
- It has high **strategic impact**. (Piercy et al. 2010)
- It can be determined in three ways
(Ingenbleek et al. 2003, Monroe 1990, Hinterhuber 2008)
 - Cost-based
 - Competition-based
 - Value-based



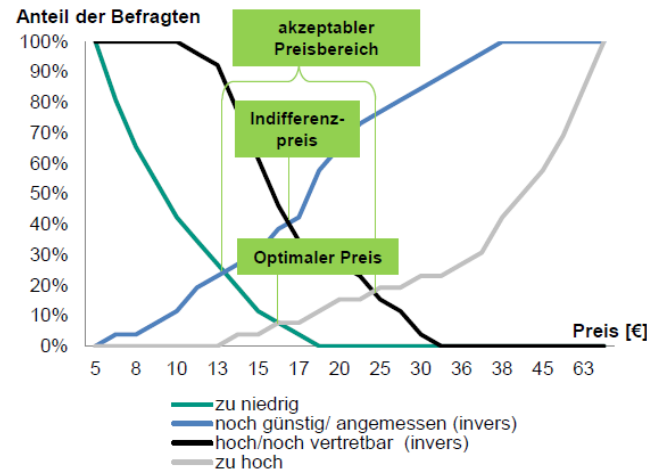
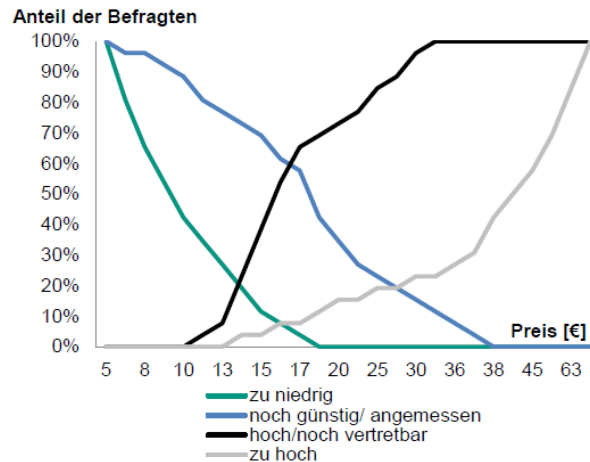
Survey: Van Westendorp-Method

Give us a price that is adequate, but still attractive

Give us a price that is high, but still acceptable

Give us the first price that you perceive as too high

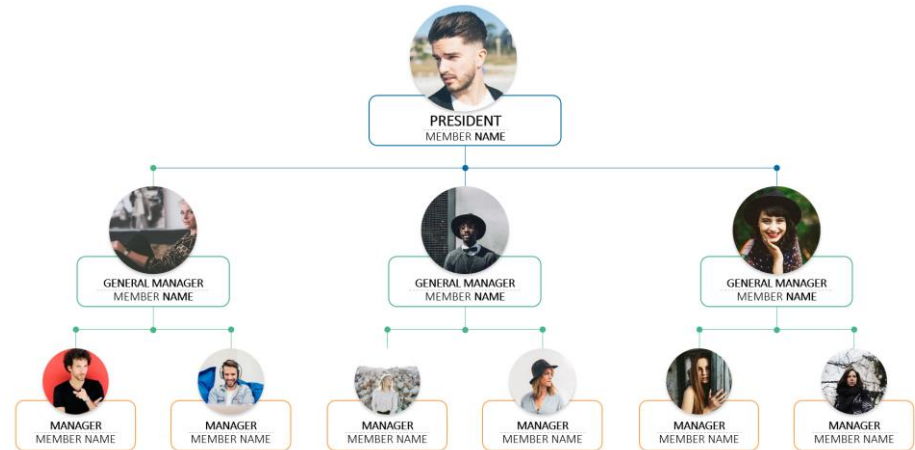
Give us the first price where you will have doubts concerning the quality, service or security of the product/service



4| Operations Plan

4.2. Firm Organization

- Planned legal entity
- Organizational chart for the firm
- ESF (European Structural Fund) objectives
 - Gender equality,
 - Non-discrimination,
 - Sustainable development



Organizational chart

Constituting a Business

Gesellschaft bürgerlichen Rechts (GbR)

- Unlimited personal liability, no capital required
- Joint management (if not agreed differently)

Gesellschaft mit beschränkter Haftung (GmbH)

- Limited personal liability, 25 k€ minimum capital
- Notarial act for partners' agreement (Gesellschaftervertrag)

Unternehmergesellschaft (UG)

- Start a GmbH with 1 EUR.
- 25% of the profits must go into a reserve until the minimum initial capital of 25 k€ has been raised.



4| Operations Plan

4.3. Risk Assessment

- Development risks for the product or service
- Risks to find adequate investors/lenders
- Risk to find the right employees
- Risks in market entry (e.g. will you find pilot customers?)
- Risks related to competition
- IP risks



Risk is a condition in which there is a **possibility of an adverse deviation from a desired outcome** that is expected or hoped for.

Vaughan, Vaughan 2008, p. 2

Risk = consequences * likelihood

Risk Consequence/Likelihood Matrix

Example: if the risk of *'delay in payment of sold goods'* has a likelihood of 'possible' (3) and its consequences are 'major' (4), then the level of this risk is 12, respectively **'high'** (Risk = consequences x likelihood = 4 x 3 = 12).

Hisrich & Ramadani 2017, p. 66

5	5	10	15	20	25
4	4	8	12	16	20
3	3	6	9	12	15
2	2	4	6	8	10
1	1	2	3	4	5
	1	2	3	4	5

Consequences

Critical	High	Moderate	Low
----------	------	----------	-----

Critical (not acceptable, threaten the survival of the company);
High (generally not acceptable, expected to cause some damage);
Moderate (acceptable, improbably to cause much damage);
Low (acceptable: very easy to be managed)

Consequence Scale

- [1]. *Irrelevant*—the risk doesn't impact changes in the company's goals and objectives
- [2]. *Minor*—the risk can be treated with existing resources
- [3]. *Moderate*—the impact of risk can be treated, but additional resources are required
- [4]. *Major*—treatment of the risk will require significant additional resources from other sectors or sources
- [5]. *Significant*—the risk might cause the company to fail achieving its goals and in some cases can prove to be fatal to the company

Likelihood Scale

- [1]. *Rare*—the risk might occur only in extraordinary circumstances. Such a risk has occurred somewhere else and might occur once in every 5+ years. Probability of occurrence is lower than 5%.
- [2]. *Unlikely*—the risk might occur at some point, for example, once in 5 years. Probability of occurrence is 5–30%.
- [3]. *Possible*—the risk might occur at some point, for example, once in 3 years. Probability of occurrence is 30–70%.
- [4]. *Likely*—the risk might occur, at least once during the year. Probability of occurrence is 70–95%.
- [5]. *Almost certain*—the risk is expected to occur in the majority of cases, occurs often during the relevant year. Probability of occurrence is 95–100%.

Options to treat risks

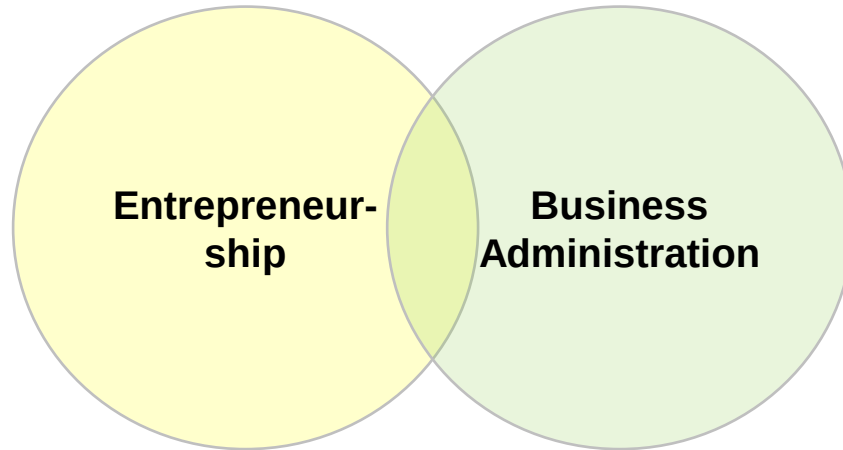
Risk avoidance includes taking proactive measures, such as requiring clients to cover purchased goods with credit through a collateral, or not undertaking any activities at all, which is expected to be damaging.

Risk reduction includes taking concrete measures to minimize the consequences of a specific risk, such as installing alarms to secure assets from eventual thefts, or installing fire alarms.

Risk anticipation, in literature, also known as the self-insurance strategy, where entrepreneurs leave aside some amount of money in order to cover damages if a risk occur.

EFFECTUATION

Entrepreneurship and Bus. Administration



In **search** for
a business model

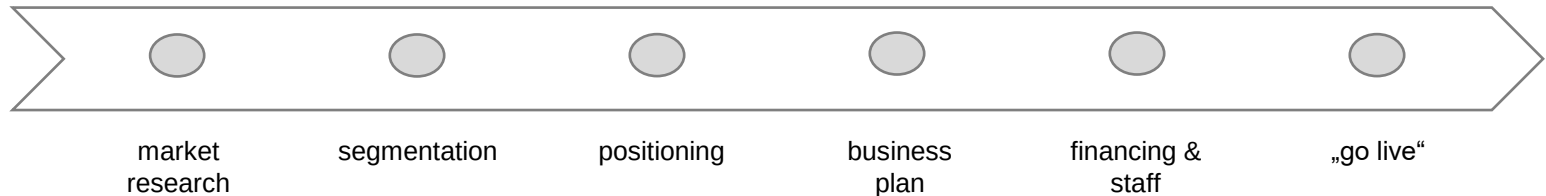
Execution of
a business model

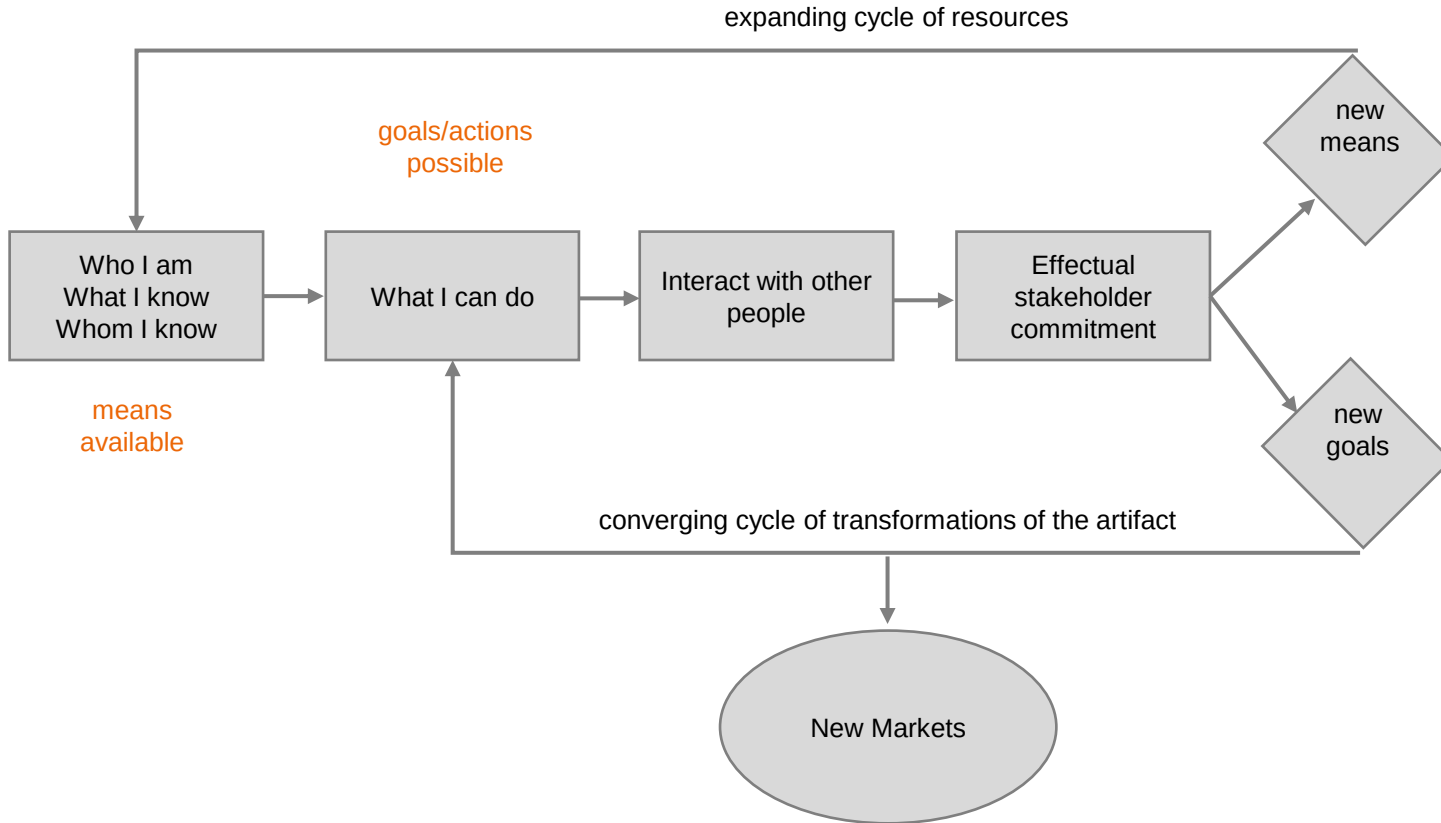
Basic Management Process

Linear causation approach



or in an entrepreneurial context ...





Effectuation Loop

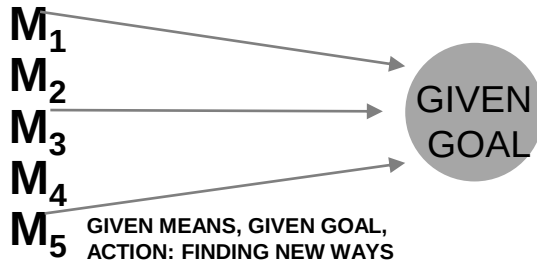
Bird-in-Hand Principle

As opposed to goal driven action, entrepreneurs take **means driven action** when building a new venture.

Based on: who I am, what I know, and whom I know.

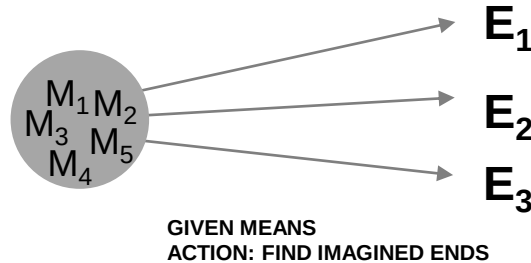
Managerial Thinking (Causal)

Selecting between given means and try to find new ways to achieve a pre-determined, "given" goal



Entrepreneurial Thinking (Effectual)

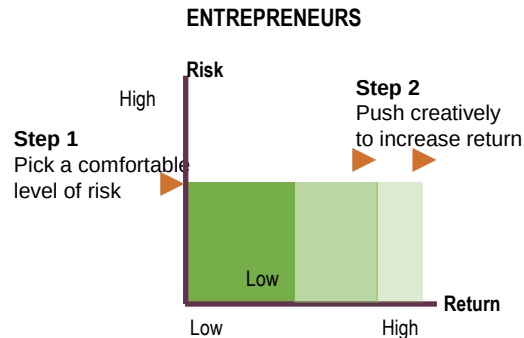
Imagining a possible new end using a given set of means



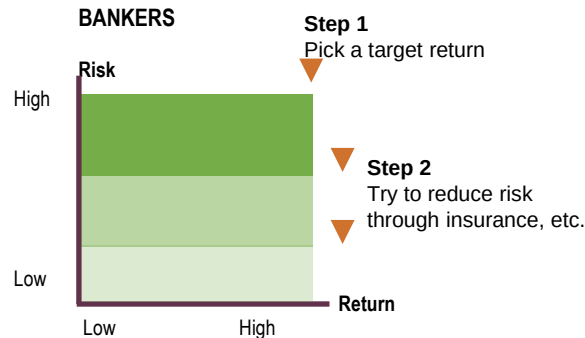
Affordable Loss Principle

As opposed to investing in calculations about expected returns, entrepreneurs limit risk by defining what they can **afford to lose** at each step.

Then goals and actions are chosen **where there is upside** even if the downside ends up happening.



For a given level of risk, entrepreneurs feel they can expand the problem space and increase returns



For a given level of return, bankers feel they can reduce the problem space and decrease risk.

Crazy Quilt Principle

- As opposed to determining the goal and then deciding who comes on board, entrepreneurs build **partnerships** with **self-selecting** stakeholders.
- By obtaining **pre-commitments** from these key partners early on in the venture, they reduce uncertainty and at the same time **co-create the new market** with its interested participants.



Lemonade Principle

- As opposed to trying to avoid or overcome surprises or adapt to them, entrepreneurs invite the **surprise factor**.
- Instead of wasting time with creating “what-if” scenarios to deal with potentially happening worst-case situations, entrepreneurs **leverage contingencies** by interpreting “bad” news and surprises as potential clues to create new markets.
- Magnetron made chocolate melt...



Pilot-in-the-Plane Principle

- As opposed to limiting entrepreneurial efforts to exploiting exogenous factors entrepreneurs know their actions will result in the **desired outcomes**.
- So they focus on **activities within their control**.
- This worldview is rooted in the belief that **the future is** neither found nor predicted, but rather **made**.



REFERENCES

References

- Ansoff, Harry Igor (1965): Corporate strategy. An analytic approach to business policy for growth and expansion. New York NY: McGraw-Hill.
- Barney, Jay (1991): Firm resources and sustained competitive advantage. In *Journal of management* 17 (1), pp. 99–120.
- Brinckmann, Jan; Grichnik, Dietmar; Kapsa, Diana (2010): Should entrepreneurs plan or just storm the castle? A meta-analysis on contextual factors impacting the business planning-performance relationship in small firms. In *Journal of Business Venturing* 25 (1), pp. 24–40.
- Bundesministerium für Wirtschaft und Technologie (BMWi). (2010). GründerZeit: Preisgestaltung. Abgerufen von http://www.dykiert-beratung.de/dyk-pdfs/dyk-file_id45-66013.pdf
- Byers, Thomas H.; Dorf, Richard C.; Nelson, Andrew J. (2011): Technology ventures. From idea to enterprise. 3. ed. New York, NY: McGraw-Hill.
- Cristea, Alexandru; Frankenberger, Karolin; Heucher, Martin; Ilar, Daniel; Kubr, Thomas; Marchesi, Heinz et al. (2016): Planen, gründen, wachsen. Mit dem professionellen Businessplan zum Erfolg. 8., aktualisierte Auflage. München: REDLINE Verlag.
- Cyr, Linda A. (2007): Creating a business plan. Expert solutions to everyday challenges. Boston, Mass.: Harvard Business School Publ (Pocket mentor series).

References

- Gruber, Marc (2007): Uncovering the value of planning in new venture creation. A process and contingency perspective. In *Journal of Business Venturing* 22 (6), pp. 782–807.
- Hatzijordanou, Nadja (2019a): Competitor Analysis Framework. Dissertation. Karlsruhe: KIT Open. Available online at <https://publikationen.bibliothek.kit.edu/1000092737>, checked on 2/20/2020.
- Hatzijordanou, Nadja (2019b): Towards Conducting Viable Competitor Analysis in Early-Stage Startups. A Design Science Approach. Karlsruher Institut für Technologie (KIT). Available online at <https://doi.org/10.1007/s11301-019-00158-5>.
- Hinterhuber, A. (2008). Customer Value-Based Pricing Strategies: Why Companies Resist. *Journal of Business Strategy*, 29(5), 41–50.
- Hisrich, Robert D.; Ramadani, Veland (2017): Effective entrepreneurial management. Strategy, planning, risk management, and organization. Cham, Switzerland: Springer.
- Homburg, Christian (2017): Grundlagen des Marketingmanagements. Einführung in Strategie, Instrumente, Umsetzung und Unternehmensführung. 5., überarbeitete und erweiterte Auflage. Wiesbaden: Springer Gabler.

References

- Ingenbleek, P., Debruyne, M., Frambach, R., & Verhallen, T. (2003). Successful new product pricing practices: A contingency approach. *Marketing Letters*, 14(4), 289–305.
- McGrath, Rita Gunther; MacMillan, Ian (1995): Discovery Driven Planning. In *Harvard Business Review* 73, pp. 44–54.
- McKeever, Mike P. (2019): How to write a business plan. 14th edition. Berkeley, California: Nolo.
- Moore, Geoffrey A. (2001): Crossing the chasm. Marketing and selling high-tech products to mainstream customers. New York: PerfectBound.
- Monroe, Kent B. (2003): Pricing. Making profitable decisions. Boston: McGraw-Hill/Irwin.
- Nag, Rajiv; Hambrick, Donald C.; Chen, Ming-Jer (2007): What is strategic management, really? Inductive derivation of a consensus definition of the field. In *Strategic management journal* 28 (9), pp. 935–955.

References

- Piercy, N. F., Cravens, D. W., & Lane, N. (2010). Thinking strategically about pricing decisions. *Journal of Business Strategy*, 31(5), 38–48.
- Porter, Michael E. (1979): How competitive forces shape strategy. In *Harvard Business Review* 57 (2), pp. 137–145.
- Porter, Michael E. (2008): The five competitive forces that shape strategy. In *Harvard Business Review* 86 (1), pp. 25–40.
- Porter, Michael E. (2008): What is strategy? In *Harvard Business Review* 86 (1), pp. 2–21.
- Sarasvathy, Saras D. (2009): *Effectuation. Elements of entrepreneurial expertise*: Edward Elgar Publishing.
- Simon, H. (2004). Ertragssteigerung durch effektivere Pricing-Prozesse. *ZfB*, 74(11), 1083–1102.
- van Westendorp, Peter H. (1976): NSS Price Sensitivity Meter (PSM)-A new approach to study consumer perception of prices. In : *Proceedings of the ESOMAR Congress*, pp. 139–167.
- Wirtz, Bernd W. (2011): *Business model management. Design - instruments - success factors*. 1. ed. Wiesbaden: Gabler.
- Vaughan, Emmett J.; Vaughan, Therese M. (2008): *Fundamentals of risk and insurance*. 10. ed. Hoboken, NJ.: Wiley.

Thank You!

